

Ahnaf Shahriar

[Email](#) | [LinkedIn](#) | [Github](#)

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

Sept. 2021 – May 2026

- Recipient of Richard & Elizabeth Madter Entrance Scholarship and President's Scholarship of Distinction
- **Relevant Courses:** Digital Communication Systems, Computer Networks, Computer Architecture, Real Time Operating Systems, Digital Hardware systems(Verilog)

EXPERIENCE

Undergraduate Research Assistant

May. 2025 – Present

University of Waterloo

Waterloo, ON

- **Research:** Conducting Case studies on Static Analysis type checking for Java.
- **Compiler Optimization:** Implementing bytecode storage on Nullness checker for EISOP framework.

Wireless Firmware Developer

Jan. 2025 – Apr. 2025

ON Semiconductor Corporation

Waterloo, ON

- **SDK Development:** Integrated *I3C* support and *BLE* profile services into sample code.
- **Wireless Connectivity:** Developed and integrated cutting-edge *LE Audio*, *HFP*, and *A2DP* functionality into an all-in-one Application.
- **Wireless Manager:** Modularized and implemented a state machine for power efficiency by *25%*
- **Test Automation:** Automated SDK test cases, reducing manual testing efforts by up to *80%*

IC Design and Verification Intern

Jan. 2024 – May 2024

NXP Semiconductors Canada

Kanata, ON

- **IP Design:** Developed multiple IP blocks for TCAM and Ethernet Parsing for NoC upto *100Gbps*.
- **Timing Analysis:** Spearheaded critical path improvements for IP to meet *600Mhz* from *400Mhz*.
- **Functional Testing:** Designed brand new End-to-End functional tests for *ECC Detection* on Chip.

IC Design Verification Intern

May 2023 – Aug. 2023

NXP Semiconductors Canada

Kanata, ON

- **UVM SystemVerilog:** Designed the *IP specific Virtual Sequence* to complete corresponding IP coverage exceeding *95%*.
- **Test Planning:** Created Simulation scenarios for testing features in High speed Dataplane processing.
- **Simulation Testing:** Created Simulation scenarios for testing features in High speed Dataplane processing.

Embedded Software Engineering Intern

Sept. 2022 – Dec. 2022

Synapse Product Development

Seattle, WA

- **Prototyping:** Leveraged Zephyr RTOS to create a proof of concept on *NRF52 BLE* device.
- **Python APIs:** Developed company specific lab automation software for equipment from *Agilent*, *Keysight*, *NI*, *Tektronik*.
- **Automation:** Streamlined testing and in house procedures using *Python* and *Bash*.
- **Driver Development:** Designed and implemented *drivers* for the controls of PCB testing Device(*I2C*, *UART*)

PROJECTS

RISC-V processor: A 5 stage pipelined FPGA processor in Verilog. Designed and tested with Vivado for Zynq-7000.

Stereo System: An FPGA designed in Quartus and programmed in C. Implements stereo playback speed options.

Real Time Executable: A STM32 RTOS implementation capable of Pre-emptive task switching and its very own Malloc.

VHDL Compiler: A Java Compiler for creating combinational VHDL circuits. Using a boolean intermediate representation

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Tcl, Perl and Bash scripting, ASM, VHDL, SystemVerilog/Verilog

Tools: Quartus, Git, Linux, Qemu, WireShark, UVM, Matlab, Synopsys DC, Cadence Xcelium, Vivado

Hardware: Oscilloscopes, Logic Analyzer, Multimeters, Spectrum Analyzer

Protocols: TCP/IP, JTAG, Serial, Ethernet, CAN/CAN-FD, LIN, AXI